

Verdier-Dual Loeb-Rosser Recollement

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Abstract

Pass 65 studies the second recollement triangle for the finite generic-point model $X_S = \{\eta\} \sqcup \{(p) : p \in S\}$ of the Loeb-Rosser dictionary. Pass 64 used the triangle $j_! j^* \rightarrow \text{id} \rightarrow i_* i^*$. This pass uses

$$i_* i^! \rightarrow \text{id} \rightarrow Rj_* j^*,$$

and checks the finite Verdier-dual form of the mixed class ϵ_S .

Main Result

The $i^!$ / local-support spine is the two-term complex

$$\mathbb{Z} \xrightarrow{\Delta} \mathbb{Z}^S, \quad 1 \mapsto (1, \dots, 1).$$

Therefore

$$H^0(i^!) = 0, \quad H^1(i^!) \cong \text{coker } \Delta \cong \mathbb{Z}^{|S|-1}.$$

So the Rosser horizontal lattice has both a $j_!$ presentation and an $i^!$ closed-support presentation.

Let

$$d_S : \mathbb{Z}^S \rightarrow \mathbb{Z}^{|S|-1}, \quad (x_p) \mapsto (x_p - x_{p_0})_{p \neq p_0}$$

represent the recollement boundary / mixed class ϵ_S . On the finite matrix spine, Verdier duality sends

$$d_S \mapsto -d_S^T,$$

so the finite functional equation is

$$\mathbb{D}(\epsilon_S) = -\epsilon_S^\vee.$$

The sign is invisible over $\mathbb{F}_2$ but is visible over $\mathbb{Z}$ as an orientation datum of the gluing triangle.

Verification

The checker `code/scripts/check-pass65.py` generated `artifacts/reports/pass65-verdier-dual-recollement-functional-equation-check.json`.

Verified facts:

- for $s = 1, \dots, 7$, $\ker \Delta = 0$ and $\text{coker } \Delta$ has rank $s - 1$;
- d_S and $-d_S^T$ have the same rank $s - 1$;
- duality squared returns d_S ;
- the sign disappears modulo 2;
- finite-prime restriction maps commute with both d_S and the dual boundary.

Remaining Gap

This is a finite Alexandrov-space computation. The honest $\text{Spec } \mathbb{Z}$ lift still requires a dualizing normalization and a calculation of the duals of $\mathbb{Z}_p/\mathbb{Z}$ and $\widehat{\mathbb{Z}}_S/\mathbb{Z}$, with product/direct-sum behavior tracked explicitly.